Got it! I'll first generate very **detailed theory** on the topics:

- **NAT (Network Address Translation)**
- **IPv6 (Internet Protocol Version 6)**
- **WAN Technologies (Wide Area Network technologies)**

Then after that, I'll generate 50 MCQs (10 easy, 30 medium, 10 hard) with answers and explanations.

Detailed Theory

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1. NAT (Network Address Translation)

Overview:

NAT is a method used in networks to translate private (local) IP addresses into public IP addresses and vice versa. It enables multiple devices on a local network to share a single public IP address to communicate with external networks like the Internet. NAT is crucial for IPv4 networks to cope with IPv4 address exhaustion.

Types of NAT:

- **Static NAT:** One-to-one mapping between a private IP and a public IP. Used when a device must be accessible from outside (e.g., a web server).
- **Dynamic NAT:** Maps a private IP to any available public IP from a pool. No fixed mapping, works on a first-come-first-served basis.
- **PAT (Port Address Translation) / NAT Overload:** Maps multiple private IP addresses to a single public IP by using different source port numbers. This is the most common NAT form used in homes/businesses.

How NAT works:

- When a device inside the LAN sends traffic to the internet, the NAT-enabled router replaces the source IP address in the packet header with its own public IP address.
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For PAT, it also changes the source port number to maintain uniqueness.

- When a reply comes back, the router uses a NAT table to translate the destination IP and port back to the original private IP and port.

Benefits:

- Conserves public IP addresses.
- Adds security by hiding internal IPs.
- Enables private IP addressing schemes.

Limitations:

- Breaks end-to-end connectivity; protocols that embed IP addresses in payload may require special handling (e.g., FTP).
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Can complicate VPN and peer-to-peer connections.

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2. IPv6 (Internet Protocol Version 6)

Why IPv6?
IPv6 is the successor to IPv4, designed to overcome IPv4's address exhaustion and provide improvements in routing, security, and configuration.

IPv6 Addressing:

- IPv6 addresses are 128 bits long (compared to 32 bits in IPv4), allowing for 2^128 unique addresses.
- Written as eight groups of four hexadecimal digits separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
- Leading zeros in each block can be omitted, and consecutive blocks of zeros can be replaced by :: once per address for brevity.

IPv6 Address Types:

- **Unicast:** One-to-one communication.
- **Multicast:** One-to-many communication.
- **Anycast:** One-to-nearest communication (packets delivered to the nearest of multiple potential receivers).
- **No Broadcast:** IPv6 eliminates broadcast; multicast replaces its function.

IPv6 Features:

- **Simplified header:** Fixed header length (40 bytes) with fewer fields for faster processing.
- **Stateless Address Autoconfiguration (SLAAC):** Devices can self-configure their IPv6 addresses automatically.
- **Built-in IPsec:** Security is mandatory for IPv6, although optional in IPv4.

- **Extensive support for multicast:** More efficient network communication.
- **No NAT needed:** Vast address space reduces need for address translation.

IPv6 Address Scopes:

- **Link-local addresses (FE80::/10):** Used within a single link or subnet, automatically assigned.
- **Global unicast addresses:** Routable on the internet.
- **Unique local addresses (ULA, FC00::/7):** Private addresses within a site.

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3. WAN Technologies (Wide Area Network)

Overview:
WAN technologies connect networks over large geographic areas. They enable communication between LANs, data centers, branch offices, and the internet.

Common WAN Technologies:

- **Leased Lines:**
  Dedicated point-to-point connection between two sites. Offers guaranteed bandwidth and low latency. Examples: T1, E1, T3, E3 lines.

- **Circuit-Switched Networks:**
  Establishes a dedicated circuit for the duration of a communication session (e.g., traditional telephone networks).

- **Packet-Switched Networks:**
  Data is broken into packets and routed independently. Includes technologies like Frame Relay, X.25, and MPLS.

- **Frame Relay:**
  A WAN protocol that provides virtual circuits over a shared network. Used widely before MPLS.

- **MPLS (Multiprotocol Label Switching):**
  Directs data from one node to the next based on short path labels rather than long network addresses. Efficient and scalable, commonly used in enterprise WANs.

- **DSL (Digital Subscriber Line):** Provides broadband internet over telephone lines.
- **Cable Modem:** Uses cable TV lines for broadband internet.
- **Satellite:** Provides WAN access where terrestrial networks are unavailable.
- **Wireless WAN:** Includes 4G/5G cellular networks for wide-area mobile connectivity.

WAN Connection Types:

- **Point-to-Point:** Direct connection between two endpoints.
- **VPN (Virtual Private Network):** Creates a secure tunnel over a public WAN.

- **Broadband:** Shared, high-speed internet access.

WAN Protocols:

- **PPP (Point-to-Point Protocol):** Used for dial-up and direct connections.
- **HDLC:** Default Layer 2 protocol for serial links on Cisco devices.
- **ATM (Asynchronous Transfer Mode):** High-speed packet switching technology, less common now